

Math 125 Final Review 1

Solutions on the following page.

1. Fall 2019 Q5
2. Spring 2019 Q7
3. Fall 2018 Q6
4. Spring 2017 Q7

1. Q1

- (a) Decreasing on $[-2, -1)$ and $(-1, 2)$, no extrema
- (b) Inflection points at $x = -1.84$ and $x = 1.38$
Concave up: $(-2, -1.84)$ and $(-1, 1.38)$
Concave down $(-1.84, -1)$ and $(1.38, 2)$
- (c) Vertical asymptote: Limit is $-\infty$ on the left, ∞ on the right
No horizontal asymptotes

2. Q2

- (a) VA: At $x = 1$, $+\infty$ on left, $-\infty$ on right
HA: $y = -1$ for $x \rightarrow \infty$, $y = 1$ for $x \rightarrow -\infty$
- (b) Decreasing for $x < -1$
Local extrema at $(-1, 2^{-3/4})$
Increasing for $-1 < x < 1$ and $1 < x$
- (c) Concave down: $(-\infty, -1.64)$ and $(1, \infty)$
Inflection point at $x = -1.64$
Concave up: $(-1.64, 1)$

3. Q3

- (a) Continuous for all real numbers
- (b) Decreasing for $x < -\sqrt{2}$
Local min. at $x < -\sqrt{2}$
Increasing for $-\sqrt{2} < x < \sqrt{2}$
Local max at $x = \sqrt{2}$
Decreasing for $x > \sqrt{2}$
- (c) Concave down for $x < -\sqrt{6}$
Concave up for $-\sqrt{6} < x < 0$
Concave down for $0 < x < \sqrt{6}$
Concave up for $x > \sqrt{6}$
IPs at $x = -\sqrt{6}, 0, \sqrt{6}$
- (d) No VAs, HAs of $y = 0$ as $x \rightarrow \pm\infty$ (make sure to evaluate the limit)

4. Q4

- (a) Compute $f'(x) = \frac{1}{2\sqrt{1+x}}$
- (b) Rearrange inequality to get $\frac{f(x)-f(1)}{x-1} \leq \frac{1}{2\sqrt{2}}$

- (c) If $\frac{f(x)-f(1)}{x-1} > \frac{1}{2\sqrt{2}}$, use MVT to get $1 \leq c \leq x$ with $f'(c) > \frac{1}{2\sqrt{2}}$
- (d) Use the fact that f is bounded/decreasing to come to a contradiction